

RYTHAM MAALPANI

Chennai, TamilNadu

☎ +91-6380165134 ✉ rythammaalpani44@gmail.com [in](#) [LinkdIn](#) [GH](#) [GitHub](#) [LeetCode](#)

Summary

Computer Science undergraduate with hands-on experience in full-stack development and database technologies. Skilled in building web applications using React.js, Node.js, and MongoDB, with working knowledge of Python for backend and data processing. Passionate about exploring emerging technologies including quantum computing concepts and their applications in database optimization. Strong problem-solving foundation demonstrated through consistent coursework performance and competitive programming practice.

Education

B.Tech Computer Science Engineering

2023-2027

Graphic Era University, Dehradun

CGPA: 8.5 / 10

DAV Boys Senior Sec School

Jun 2019 – May 2021

PCM - Class XII

Percentage: 89%

Skills

Languages: Java, JavaScript, Python

Web Development: HTML5, CSS3, React.js, Redux, Tailwind CSS, Bootstrap

Backend: Node.js, Express.js, Flask, REST APIs

Databases: MySQL, MongoDB

Tools: Docker, Git, GitHub **Core Concepts:** Data Structures and Algorithms, Machine Learning, System Design

Projects

After the Last Page | *Node.js, React, Express, MongoDB, Rest APIs*

Jan 2026 – Currently

- **Built a full-stack platform** that allows readers to create and participate in structured discussions centered around books.
- Designed a reader-focused interface optimized for long-form reading with a clear information hierarchy to improve readability.
- Implemented a scalable thread architecture supporting dynamic creation and navigation of book-specific discussions.
- Developed modular frontend components and backend REST APIs to manage discussion content and threads.

Customer Churn Prediction Platform | *Pandas, Scikit-learn, XGBoost*

February 2025 – March 2025

- Built an end-to-end churn prediction pipeline that converts telecom customer data into predictive insights for retention strategies.
- Trained an XGBoost model achieving **85 percent accuracy** using engineered behavioral features and SMOTE to address class imbalance.
- Created an interactive Tableau dashboard visualizing churn probability, customer segmentation, and revenue impact.

Quantum-Enhanced Query Optimizer | *Python, Quiskit, Scikit-learn, XGBoost, SQL*

February 2025 – March 2025

- Pioneered research combining quantum computing with database query optimization, developing hybrid classical-quantum framework for complex join order optimization.
- Engineered QUBO-based encoding scheme for D-Wave quantum systems, achieving 40% faster convergence to optimal query plans compared to traditional optimizers.

Achievements

- **800+ Data Structures and Algorithms problems** across multiple coding platforms with a peak rating of **1900**
- **99.6 percentile in Mathematics** in JEE Main, demonstrating strong analytical and quantitative problem-solving ability.